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Practical No. 1

Aim: Using the software like JDK 1.8, Eclipse IDE, Apache tomcat server 7.0 Servlets, Spring framework design and develop Web applications using MVC Framework

Implementation:

1. Open Eclipse IDE and create a new Maven project
2. Select the catalog as internal and below that select the last option of the list mentioning 'web_app' and click 'Next'
3. Enter the group_id: com.udcs and artifact_id: firstwebapp

New Maven project

Specify Archetype parameters



Group Id:	com.udcs
Artifact Id:	firstwebapp
Version:	0.0.1-SNAPSHOT
Package:	com.udcs.firstwebapp

4. Open pom.xml and edit to match the following

<project

xmlns="http://maven.apache.org

g/POM/4.0.0"

xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

xsi:schemaLocation="http://maven.apache.org/POM/4.0.0

http://maven.apache.org/maven-v4_0_0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.gjccs</groupId>

<artifactId>FirstWebApp</artifactId>

<packaging>war</packaging>

<version>0.0.1-SNAPSHOT</version>

<name>FirstWebApp Maven Webapp</name>

<properties>

<jdk.version>1.7</jdk.version>

<spring.version>3.2.13.RELEASE</spring.version>

<jstl.version>1.2</jstl.version>

```

    </properties>
    <dependencies>
      <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-webmvc</artifactId>
        <version>${spring.version}</version>
      </dependency>
      <dependency>
        <groupId>javax.servlet</groupId>
        <artifactId>jstl</artifactId>
        <version>${jstl.version}</version>
      </dependency>
    </dependencies>
    <build>
      <plugins>
        <plugin>
          <groupId>org.apache.maven.plugins</groupId>
          <artifactId>maven-compiler-
            plugin</artifactId>
          <version>3.3</version>
          <configuration>
            <source>${jdk.version}</source>
            <target>${jdk.version}</target>
          </configuration>
        </plugin>
        <plugin>
          <groupId>org.apache.maven.plugins</groupId>
          <artifactId>maven-war-plugin</artifactId>
          <version>3.3.1</version>
        </plugin>
        <plugin>
          <groupId>org.eclipse.jetty</groupId>
          <artifactId>jetty-maven-plugin</artifactId>

```

```

<version>9.2.11.v20150529</version>

<configuration>

    <scanIntervalSeconds>10</scanIntervalSeconds>

</webApp>
<contextPath>/spring3</contextPath>
</webApp>
</configuration>

</plugin>
<plugin>

    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-eclipse-plugin</artifactId>

</version>2.9</version>
<configuration>
<downloadSources>true</downloadSources>
    <downloadJavadocs>true</downloadJavadocs>
    <wtpversion>2.0</wtpversion>
    <wtpContextName>spring3</wtpContextName>
</configuration>

</plugin>
</plugins>
</build>

</project>

```

5. Go to web.xml in src/main/webapp/web-inf and edit it as shown below

```

<web-app
    xmlns="http://java.sun.com/xml/ns/javaee"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://java.sun.com/xml/ns/javaee
        http://java.sun.com/xml/ns/javaee/web-app_2_5.xsd" version="2.5">
    <display-name>MVC Application</display-name>
    <servlet>
        <servlet-name>spring-web</servlet-name>
        <servlet-class>org.springframework.web.servlet.DispatcherServlet </servlet-class>

```

```

        <load-on-startup>1</load-on-startup>
        <!--
<init-param><param-name>contextConfigLocation</param-name><param-
value>/WEB-
INF/spring-mvc-config.xml</param-value></init-param>
-->
    </servlet>
    <servlet-mapping>
        <servlet-name>spring-web</servlet-name>
        <url-pattern>/</url-pattern>
    </servlet-mapping>
</web-app>

```

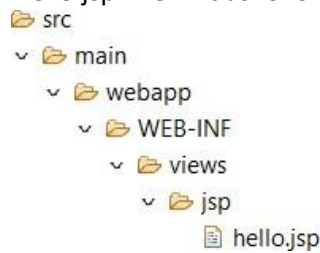
6. Right click on web-inf and create XML file named 'spring-web-servlet.xml' and edit as follows

```

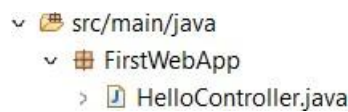
<beans
    xmlns="http://www.springframework.org/schema/b
eans"
    xmlns:context="http://www.springframework.org/sc
hema/context"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"
    xmlns:mvc="http://www.springframework.org/sche
ma/mvc"
    xsi:schemaLocation="
http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans-3.2.xsd
http://www.springframework.org/schema/mvc
http://www.springframework.org/schema/mvc/spring-mvc-3.2.xsd
http://www.springframework.org/schema/context
http://www.springframework.org/schema/context/spring-context-3.2.xsd">
    <context:component-scan base-package="com.udcs" />
    <bean
class="org.springframework.web.servlet.view.InternalResourceViewResolver">
        <property name="prefix">
<value>/WEB-INF/views/</value>
        </property>
        <property name="suffix">
            <value>.jsp</value>
        </property>
    </bean>
    <mvc:resources mapping="/resources/**" location="/resources/" />
    <mvc:annotation-driven />
</beans>

```

7. Create new folder in WEB-INF as 'views' and a subfolder as 'jsp' and create a 'hello.jsp' file in it as follows



8. Right click on src/main/resources in the file explorer and create new folder named 'java' but select 'main' in the folder creating wizard before that
9. Create a java class in java folder named 'hellocontroller.java' as shown below



10. Edit HelloController.java as follows import java.util.regex.Matcher; import java.util.regex.Pattern;

```
public class
    RegexMatches { public
    static void main(String
    args[]) {

        // String to be scanned to find the pattern.
        String line = "This order was placed for QT3000! OK?";
        String pattern = "(.*)\\d+(.*)";

        // Create a Pattern object
        Pattern r = Pattern.compile(pattern);

        // Now create
        matcher object.
        Matcher m =
        r.matcher(line);
        if (m.find()) {
            System.out.println("Found value: " + m.group(0));
            System.out.println("Found value: " + m.group(1));
            System.out.println("Found value: " + m.group(2));
        } else {
            System.out.println("NO MATCH");
        }
    }
}
```

11. Edit the 'hello.jsp' file as follows
 mySelection =
 app.activeDocument.selection;

```

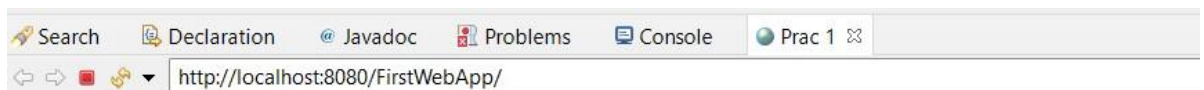
myDoc = app.activeDocument;
if (mySelection instanceof Array)
{
    selSwatches = myDoc.swatches.getSelected();

    if(selSwatches.length != 0)
        for (i=0; i<mySelection.length; i++)
        {
            if(mySelection[i].typename == "PathItem" ||
                mySelection[i].typename == "CompoundPathItem")
            {
                selItem = mySelection[i];
                selItem.filled = true;
                swatchIndex =
                    Math.round(
                        Math.random() *
                        (selSwatches.length - 1
                        ));

                if(selItem.typename == "PathItem")
                    selItem.fillColor =
                        selSwatches[swatchIndex].color;
                else
                    selItem.pathItems[0].fillColor =
                        selSwatches[swatchIndex].color;
            }
        }
    }
}

```

12. In IDE, right click on firstwebapp > Maven > update Project > check on 'force update snapshot' > Finish
13. Right click on project > run as > Maven build
14. In the following pop-up, enter the goals as 'clean install'
15. Right click on project > Run as > Run on server > Apache Tomcat v9 > select any file in configured > Finish To see the output



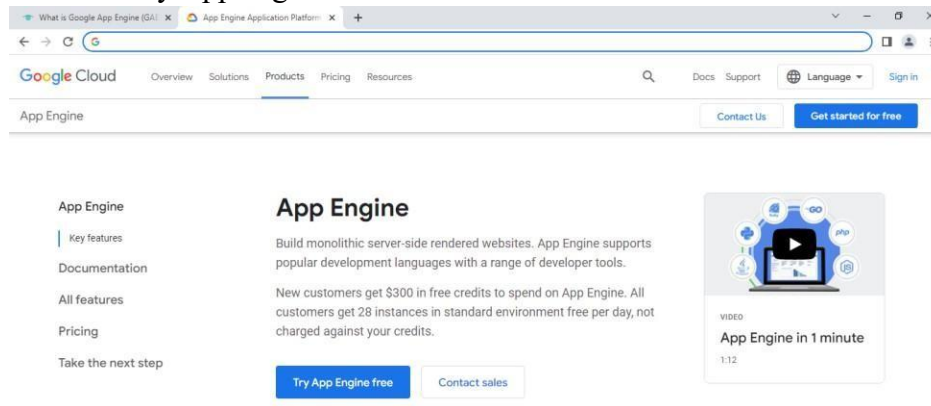
Practical No. 2

Aim: Installing and configuring the required platform for Google App Engine

Implementation:

A. Making Google App Engine account

1. Open your google account and go to the following link <https://cloud.google.com/appengine>
2. Click on Try App engine free button



3. This page will appear, add country and choose other in describes and click Continue.

Country

India

What best describes your organization or needs?

Please select

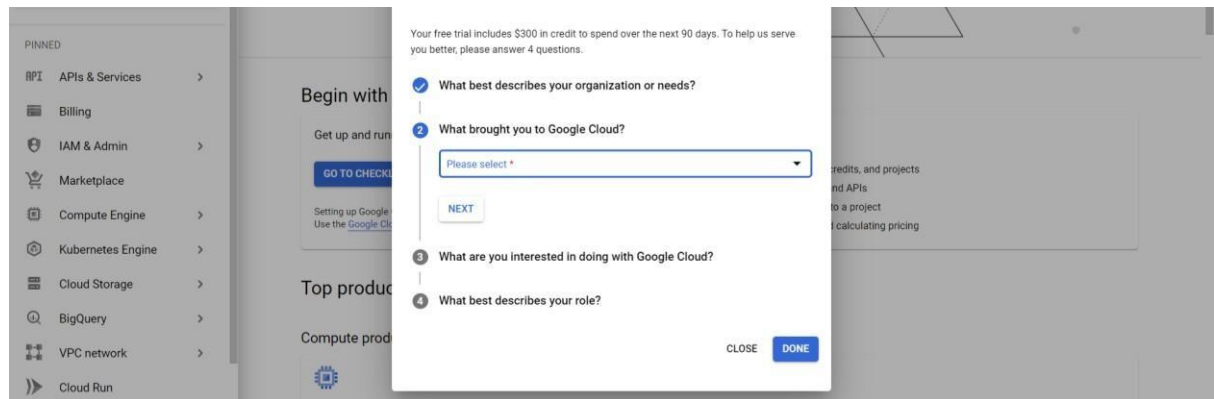
Terms of Service

☐ I have read and agree to the [Google Cloud Platform Terms of Service](#), [Supplemental Free Trial Terms of Service](#), and the terms of service of any applicable services and APIs.

Required to continue

CONTINUE

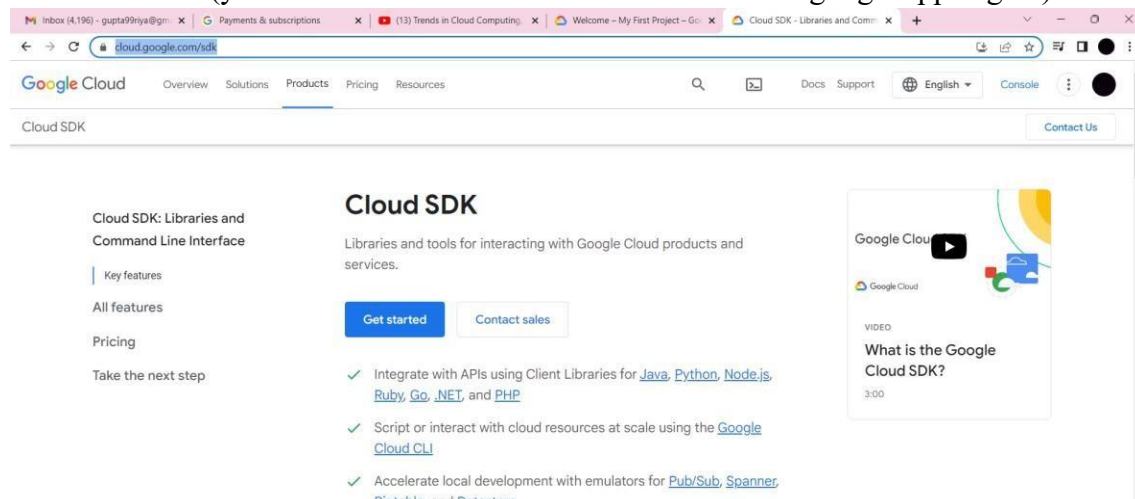
4. Add you card details (Visa or MasterCard only) and do the payment of Rs. 2
Fill this according to you purpose



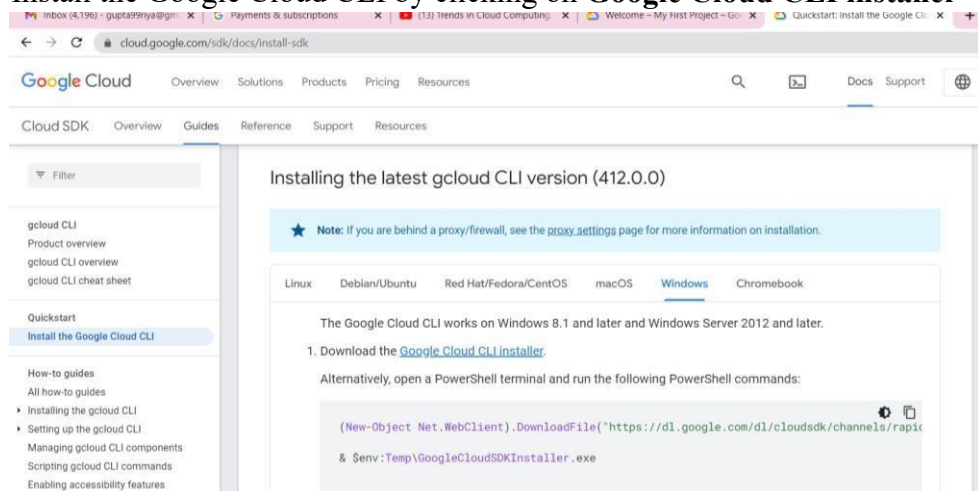
After all procedure we will get,

B. Using GAE account, to download Google Cloud SDK.

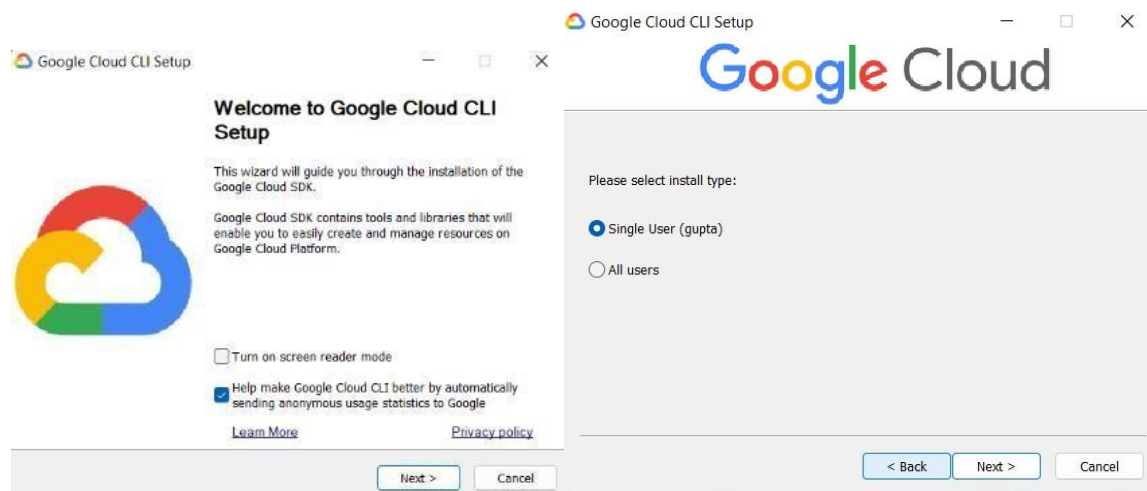
- Go to google and search **google cloud sdk** go to site <https://cloud.google.com/sdk> then Get Started (you should use this with same account used in google app engine)



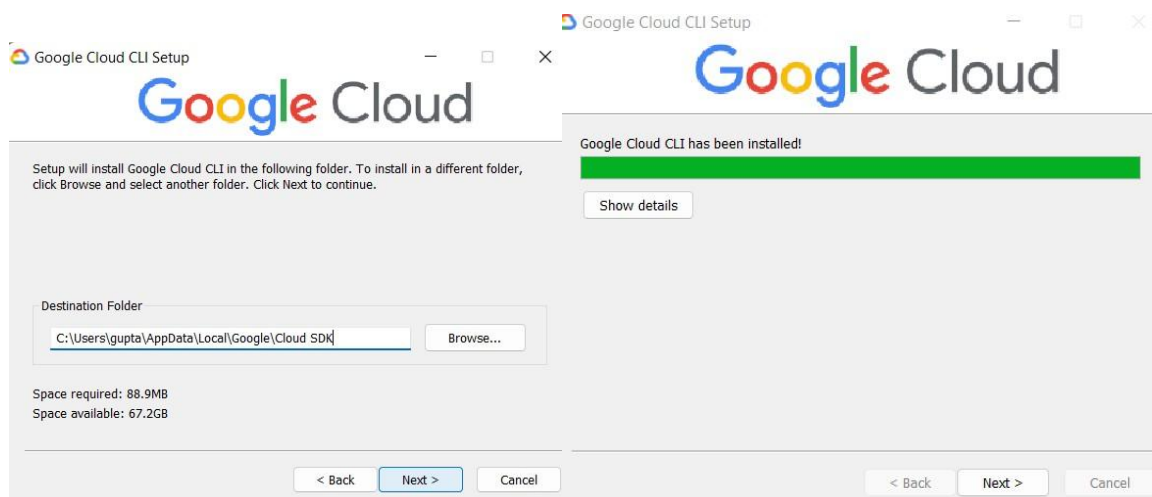
- Install the Google Cloud CLI by clicking on **Google Cloud CLI installer**



7. Launch the installer and follow the prompts. The installer is signed by Google LLC.

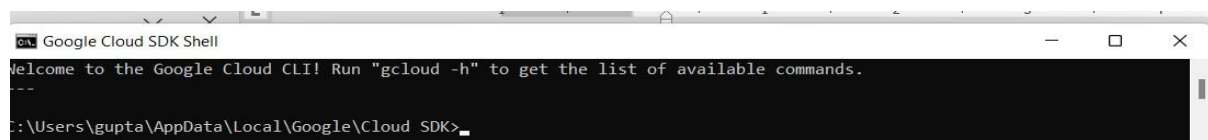


8. After installing it will ask for log in as shown.



Select 5 from it, close this and open Google Cloud SDK Shell it will look like Add the installer path in system also.

Run command ‘ gcloud components install app-engine-java’



Practical No. 3

Aim: Studying the features of the GAE PaaS model.

Platform as a Service (PaaS) - What is it?

PaaS is a category of cloud computing services that provide a computing platform and a solution stack as a service.

Along with software as a service (SaaS) and infrastructure as a service (IaaS), it is a service model of cloud computing. In this model, the consumer creates the software using tools and/or libraries from the provider/vendor. The consumer also controls software deployment and configuration settings.

The provider provides the networks, servers, storage and other services. PaaS offerings facilitate the deployment of applications without the cost and complexity of buying and managing the underlying hardware and software and provisioning hosting capabilities.

PaaS Key Features

1. Services to develop, test, deploy, host and maintain applications in the same integrated development environment
2. Web-based management/administration consoles
 - Reducing the need for system administration/dev ops
 - Resource utilization monitoring capabilities
 - Easily identify bottlenecks
 - Multi-tenant architecture
 - Certain PaaS offerings attempt to support use of the application by many concurrent users, by providing concurrency management, scalability, fail-over and security
 - Support for development team collaboration
 - Pay for what you use billing model

PaaS - Popular offerings

Heroku

One of the first cloud platforms, has been in development since June 2007, when it supported only the Ruby programming language, but has since added support for Java, Node.js, Scala, Clojure, Python and (undocumented) PHP

Windows Azure

Microsoft's cloud computing platform used to build, deploy and manage applications through a global network of Microsoft- managed datacenters

dotCloud

Founded in 2008 by Solomon Hykes, dotCloud is the first application platform designed from the ground up for modern service-oriented development

Cloud Foundry

- Developed by VMware released under the terms of the Apache License 2.0
- Primarily written in Ruby
- AppCloud runs on Cloud Foundry
- Since it is open sourced, ActiveState has created a commercial distribution of the Cloud Foundry **software for enterprises to host their own private PaaS**

Engine Yard

A San Francisco, California based, privately held platform as a service company focused on Ruby on Rails and PHP, and recently announced support for Node.js deployment and management

Google App Engine (often referred to as GAE or simply App Engine, and also used by the acronym GAE/J)

- A cloud computing platform for developing and hosting web applications in Googlemanaged data centers
- Applications are sandboxed and run across multiple servers
- Offers automatic scaling for web applications-as the number of requests increases for an application, App Engine automatically allocates more resources for the web application to handle the additional demand
- Is free up to a certain level of consumed resources. Fees are charged for additional storage, bandwidth, or instance hours required by the application
- First released as a preview version in April 2008, and came out of preview in September 2011

What is Google App Engine?

Google App Engine lets you run web applications on Google's infrastructure. App Engine applications are easy to build, easy to maintain, and easy to scale as your traffic and data storage needs grow. With App Engine, there are no servers to maintain: You just upload your application, and it's ready to serve your users.

The Application Environment

Google App Engine makes it easy to build an application that runs reliably, even under heavy load and with large amounts of data. App Engine includes the following features;

- Dynamic web serving, with full support for common web technologies
- Persistent storage with queries, sorting and transactions
- Automatic scaling and load balancing
- APIs for authenticating users and sending email using Google Accounts
- A fully featured local development environment that simulates Google App Engine on your computer

Your application can run in one of three runtime environments: the Go environment, the Java environment, and the Python environment, which gives you a choice of Python 2.5 or Python

2.7.

Why App Engine?

Pros

- Easy to Get Started
- Automatic Scalability
- The Reliability, Performance, and Security of Google's Infrastructure
- Costs less
- There is a generous free usage quota and you only pay for what you use

Cons

- Sandboxed environment limits the scope of your application
- Although we can pay for certain additional resources, there are some that have a hard limit

Traditional Way

1. Write your code
2. Configure & Deploy Web server (Apache/Tomcat)
3. Configure & Deploy SQL database
4. Maintain all of these infrastructure
5. Cost of building and maintaining the infrastructure

App Engine Way

1. Write your code
2. A set of simple configurations to let App Engine know how to serve your application

Tools Bundled with the SDK

Development Server

Uploading and

Managing an App

Uploading and

Downloading Data

ProtoRPC webapp

Framework

Local Unit Testing

Appstats

Included Libraries
(Python 2.5)

- Django, PyCrypto, YAML, zipimport

Included Libraries

(Python 2.7)

- Jinja2, PIL, webapp2, etc QBurst I meet.google.com is sharing your screen.

Practical No. 4

Aim: Creating and running Web applications (Guest book, MVC) on localhost and deploying the same in Google App Engine

Implementation:

1. Open Eclipse IDE and create a new Maven project
2. Select the catalog as internal and below that select the last option of the list mentioning 'web_app' and click 'Next'
3. Enter the group_id: com.udcs and artifact_id: firstwebapp

New Maven project

Specify Archetype parameters



Group Id:	com.udcs	▼
Artifact Id:	firstwebapp	▼
Version:	0.0.1-SNAPSHOT	▼
Package:	com.udcs.firstwebapp	▼

4. Open pom.xml and edit to match the following

<project

xmlns="http://maven.apache.or

g/POM/4.0.0"

xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

xsi:schemaLocation="http://maven.apache.org/POM/4.0.0

http://maven.apache.org/maven-v4_0_0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.gjccs</groupId>

<artifactId>FirstWebApp</artifactId>

<packaging>war</packaging>

<version>0.0.1-SNAPSHOT</version>

<name>FirstWebApp Maven Webapp</name>

<properties>

<jdk.version>1.7</jdk.version>

<spring.version>3.2.13.RELEASE</spring.version>

```

        <jstl.version>1.2</jstl.version>
    </properties>
    <dependencies>
        <dependency>
            <groupId>org.springframework</groupId>
            <artifactId>spring-webmvc</artifactId>

            <version>${spring.version}</version>
        </dependency>
        <dependency>
            <groupId>javax.servlet</groupId>
            <artifactId>jstl</artifactId>
            <version>${jstl.version}</version>
        </dependency>
    </dependencies>
    <build>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <
                <artifactId>maven-compiler-
                plugin</artifactId>
<version>3.3</version>
            <configuration>
                <source>${jdk.version}</source>
                <target>${jdk.version}</target>
            </configuration>
        </plugin>
        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <
            <artifactId>maven-war-plugin</artifactId>
<version>3.3.1</version>
        </plugin>
        <plugin>

```

```

        <groupId>org.eclipse.jetty</groupId>
        <artifactId>jetty-maven-plugin</artifactId>

        <version>9.2.11.v20150529</version>

    </configuration>

        <scanIntervalSeconds>10</scanIntervalSeconds>

    </webApp>

    <contextPath>/spring3</contextPath>
</webApp>

</configuration>

        </plugin>

        <plugin>

            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-eclipse-plugin</artifactId>

            <version>2.9</version>

            <configuration>

                <downloadSources>true</downloadSources>

                <downloadJavadocs>true</downloadJavadocs>
                <wtpversion>2.0</wtpversion>

                <wtpContextName>spring3</wtpContextName>

            </configuration>

        </plugin>

    </plugins>

</build>

</project>

```

5. Go to web.xml in src/main/webapp/web-inf and edit it as shown below

```

<web-app
    xmlns="http://java.sun.com/xml/ns/javaee"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://java.sun.com/xml/ns/javaee
        http://java.sun.com/xml/ns/javaee/web-app_2_5.xsd" version="2.5">
    <display-name>MVC Application</display-name>
    <servlet>
        <servlet-name>spring-web</servlet-name>

```

```

        <servletclass>org.springframework.web.servlet.DispatcherServlet</servlet-class>
        <load-on-startup>1</load-on-startup>
        <!--
<init-param><param-name>contextConfigLocation</param-name><param-
value>/WEB-
INF/spring-mvc-config.xml</param-value></init-param>
-->
    </servlet>
    <servlet-mapping>
        <servlet-name>spring-web</servlet-name>
        <url-pattern>/</url-pattern>
    </servlet-mapping>
</web-app>

```

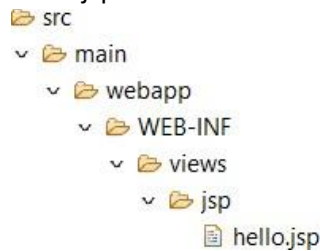
6. Right click on web-inf and create XML file named 'spring-web-servlet.xml' and edit as follows

```

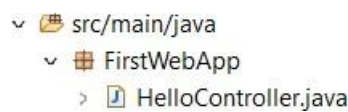
<beans
    xmlns="http://www.springframework.org/schema/b
eans"
    xmlns:context="http://www.springframework.org/sc
hema/context"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"
    xmlns:mvc="http://www.springframework.org/sche
ma/mvc"
    xsi:schemaLocation="
http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans-3.2.xsd
http://www.springframework.org/schema/mvc
http://www.springframework.org/schema/mvc/spring-mvc-3.2.xsd
http://www.springframework.org/schema/context
http://www.springframework.org/schema/context/spring-context-3.2.xsd">
    <context:component-scan base-package="com.udcs" />
    <bean
class="org.springframework.web.servlet.view.InternalResourceViewResolver">
        <property name="prefix">
<value>/WEB-INF/views/</value>
        </property>
        <property name="suffix">
            <value>.jsp</value>
        </property>
    </bean>
    <mvc:resources mapping="/resources/**" location="/resources/" />
    <mvc:annotation-driven />
</beans>

```

7. Create new folder in WEB-INF as 'views' and a subfolder as 'jsp' and create a 'hello.jsp' file in it as follows



8. Right click on src/main/resources in the file explorer and create new folder named 'java' but select 'main' in the folder creating wizard before that
9. Create a java class in java folder named 'hellocontroller.java' as shown below



10. Edit HelloController.java as follows import java.util.regex.Matcher; import java.util.regex.Pattern;

```

public class
    RegexMatches { public
    static void main(String
    args[]) {

        // String to be scanned to find the pattern.
        String line = "This order was placed for QT3000! OK?";
        String pattern = "(.*)(\\d+)(.*)";

        // Create a Pattern object
        Pattern r = Pattern.compile(pattern);

        // Now create
        matcher object.
        Matcher m =
        r.matcher(line);
        if (m.find()) {
            System.out.println("Found value: " + m.group(0));
            System.out.println("Found value: " + m.group(1));
            System.out.println("Found value: " + m.group(2));
        } else {
            System.out.println("NO MATCH");
        }
    }
}

```

11. Edit the 'hello.jsp' file as follows

```

mySelection =
app.activeDocument.selection;
myDoc = app.activeDocument;
if (mySelection instanceof Array)
{
    selSwatches = myDoc.swatches.getSelected();

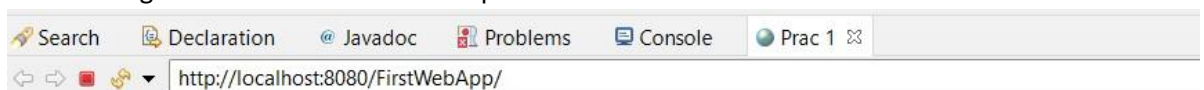
    if(selSwatches.length != 0)
        for (i=0; i<mySelection.length; i++)
        {
if(mySelection[i].typename == "PathItem" ||
    mySelection[i].typename == "CompoundPathItem")
        {
            selItem = mySelection[i];
            selItem.filled = true;
            swatchIndex =
            Math.round(
            Math.random() *
            (selSwatches.length - 1
            ));

            if(selItem.typename == "PathItem")
            selItem.fillColor =
            selSwatches[swatchIndex].color;
            else
            selItem.pathItems[0].fillColor =
            selSwatches[swatchIndex].color;

        }
    }
}

```

12. In IDE, right click on firstwebapp > Maven > update Project > check on 'force update snapshot' > Finish
13. Right click on project > run as > Maven build
14. In the following pop-up, enter the goals as 'clean install'
15. Right click on project > Run as > Run on server > Apache Tomcat v9 > select any file in configured > Finish To see the output



Hello World!

7 Guest Book:

FirstMVCWebApp Home Manual Controller

Enter Your Name:
Sachin

Enter Your Age:
35

Enter Your Contact Number:
7896541235

Enter Your Email:
sca@rdnc.ac.in

Submit

© 2026 - FirstMVCWebApp - [Privacy](#)

Privacy Policy - FirstMVCWebApp Home Manual Controller

Privacy Policy

Name = Sachin

Age = 35

Contact No = 7896541235

Email ID = sca@rdnc.ac.in

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Practical No. 5

Aim: Developing an ASP.NET based web application on the Azure platform

Implementation:

1. Go to the Azure portal and click on free account at top right of the home page and go to start page
2. Create a new account and enter debit card details to finalize the account generation process
3. You can view the following image once done



4. Make sure to have installed Visual Studio 2022, and open it
5. Create a new project > ASP.NET Core Web App > Next > Create
6. In your File tree, open pages folder and edit the index.html like so

```
<div class="text-center">
  <h1 class="display-4">UDCS Trends in Cloud Computing</h1>
  <p>Learn about <a href="https://docs.microsoft.com/aspnet/core">Practical no. 5, Hosting a ASP.NET App on Azure</a>.</p>
</div>
```

7. Run in localhost to verify its working

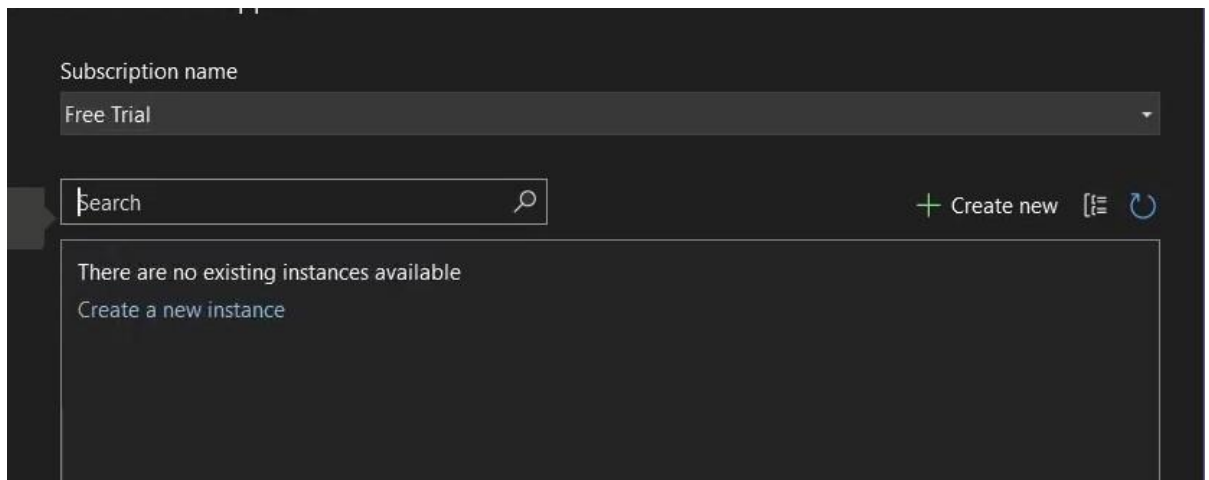
Prac5App Home Privacy

UDCS Trends in Cloud Computing

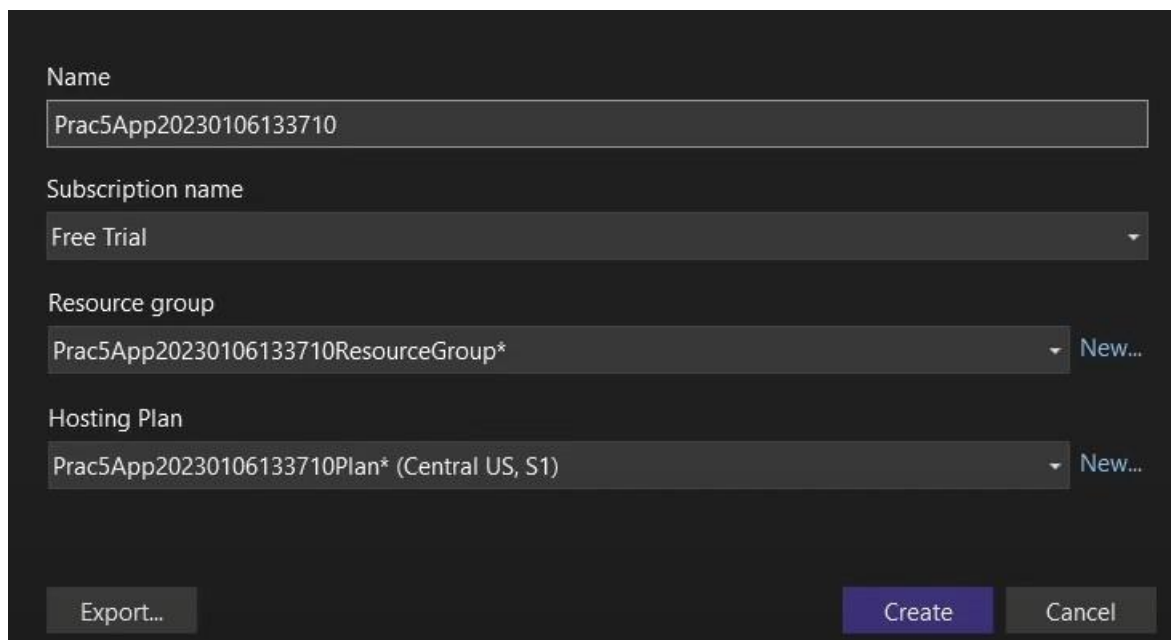
Learn about [Practical no. 5, Hosting a ASP.NET App on Azure](https://docs.microsoft.com/aspnet/core).

8. Now in Build tab, click on public [AppName]
9. Select 'Azure' > Next > Azure App Service (windows) > Next

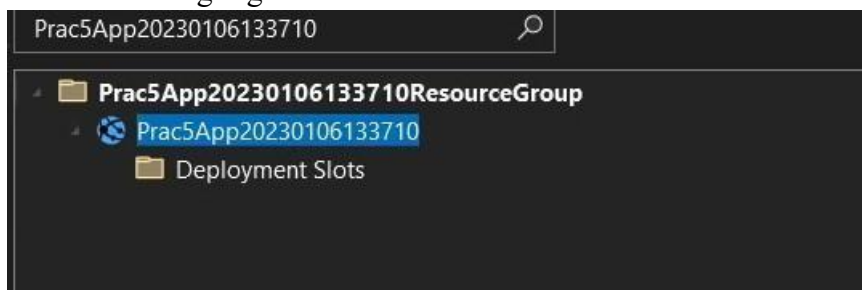
10. Make sure your Microsoft account used for Azure is signed in and currently selected
11. Go for free trial and click on the '+ Create new' to create a new resource



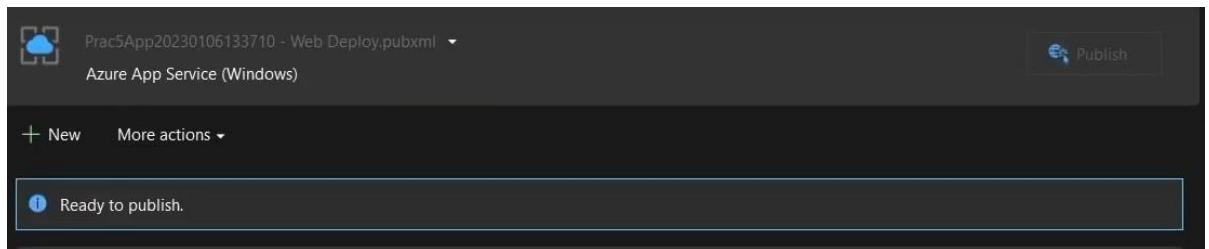
12. Click on create



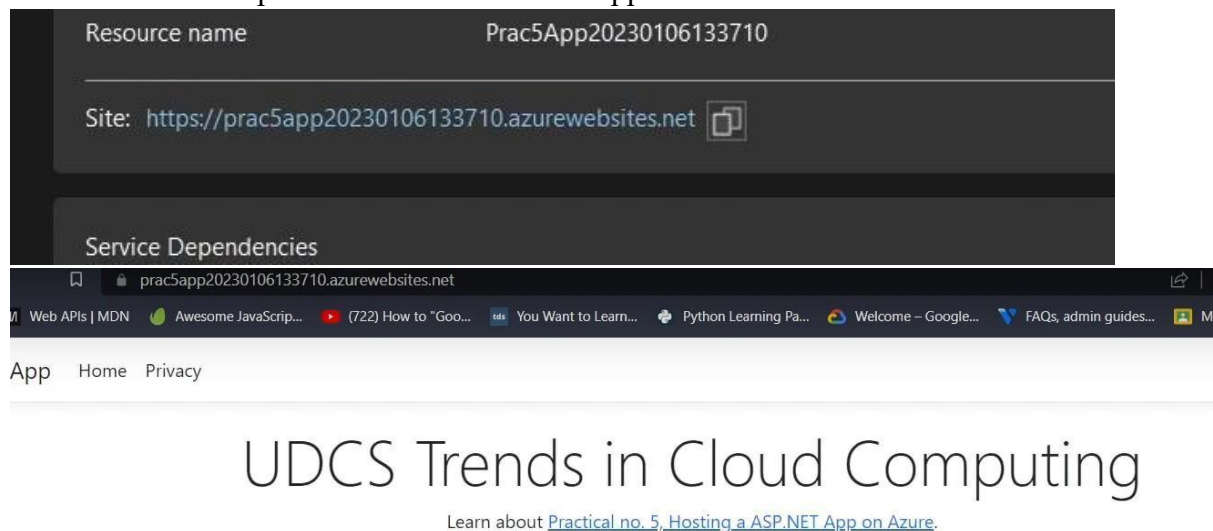
13. Click on the highlighted resource and click on Finish



14. Click on close and then click on publish



15. Scroll down to find and copy the URL generated and paste it in a browser to view the published and hosted web app

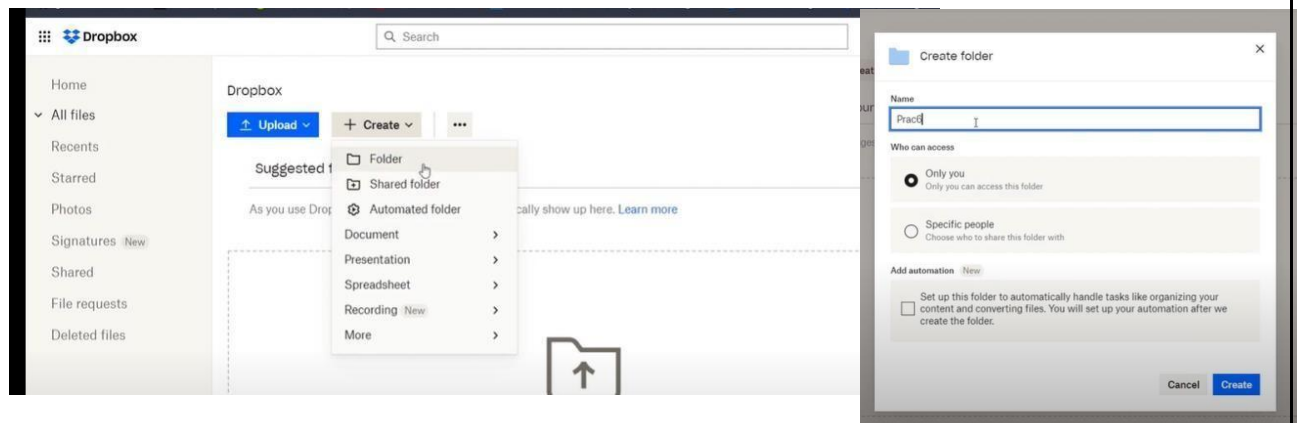


Practical 6

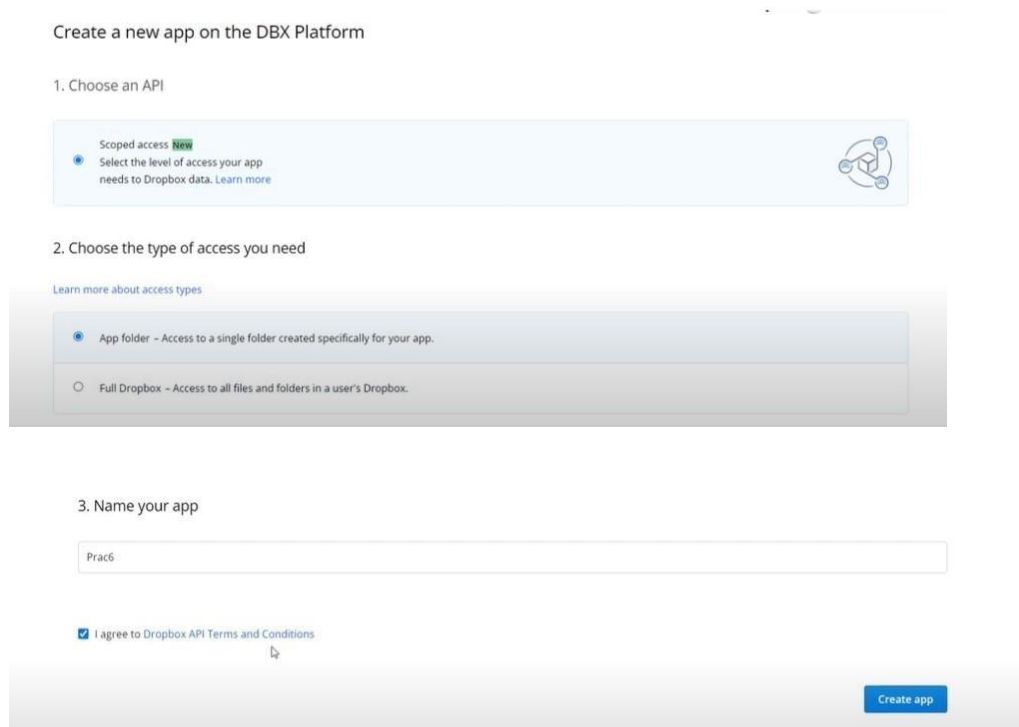
Aim: Creating an application in Dropbox to store data securely. Develop a source code using Dropbox API for updating and retrieving files.

Implementation:

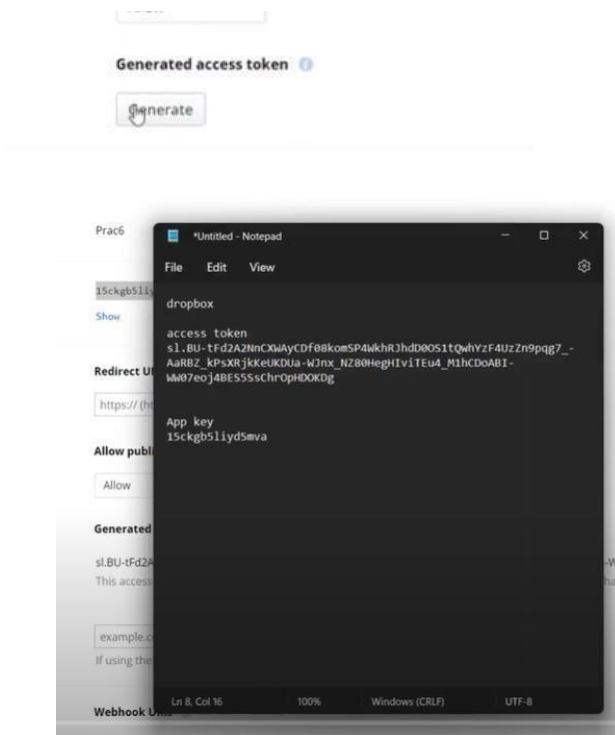
1. Go to <https://www.dropbox.com/> and create an account. You can sign in with your google account.
2. Click on [Continue with 2 GB Dropbox Basic plan](#) at the end of the page. Create a 'Prac6' folder.



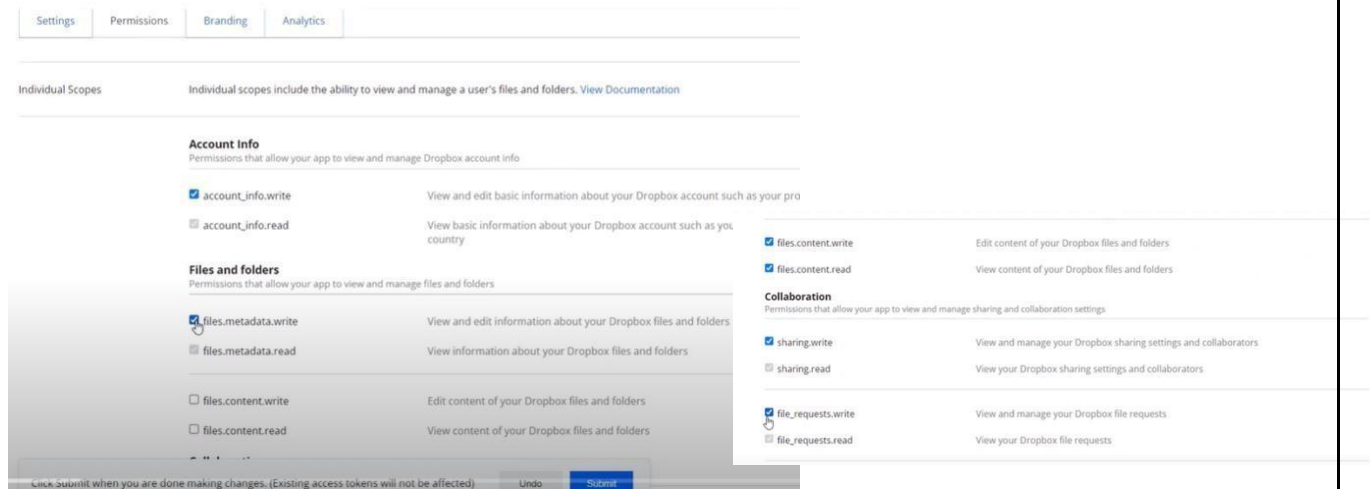
3. Go to <https://www.dropbox.com/lp/developers> and click on **Create apps**.



4. Click on generate, copy the generated access token and App key into a notepad file for later use.



5. Go to permissions and check all the following boxes. After that click on submit.



☒ contacts.write View and manage your manually added Dropbox contacts

☐ contacts.read View your manually added Dropbox contacts

OpenID Scopes

Scopes used for OpenID Connect.

At this time, team-scoped apps **cannot** request OpenID Connect scopes.

OpenID scopes must be explicitly set in the "scope" parameter on /oauth2/authorize to be requested.

Connect

Permissions that allow your app to access user login info

☒ profile Read basic profile info

☐ openid Required for OpenID Connect flow

☒ email Read basic email info

6. Go to <https://colab.research.google.com/>, create a new notebook and write the following code. Create a notepad file with some texts, upload it to dropbox folder 'Prac6'.

```
import dropbox
dropbox_access_token=
'sl.BUTEvih4dpk6Cma3FXLv2ToJfIGwWDJ6WmruJdbLeDFVLGqoE7g_Jcmy2
Yfujqz_eHH
ORr82G0gWDIASZUQgDo6co4WrbN-
YiuG5JgKiOGBRc7WlanGHglTejJhuqZ8LMLxdaWq9gvRF' #Enter your own
access token dropbox_path= "/Prac6/aa.txt"
computer_path="aa.txt"
client =
dropbox.Dropbox(dropbox_access_token
) print("[SUCCESS] dropbox account
linked")

client.files_upload(open(computer_path,
"rb").read(),
dropbox_path) print("[UPLOADED] {}".format(computer_path))
```

```
import dropbox
dropbox_access_token = 'sl.BUSshjwPUa69LP3K2PH5xJNud6pCfptseLcKhtLMB4jeLVQBOTqSLrmyFATDDOPZKEL4YeJcx-9zM3WtyR4_zxJMb2S4nDkhNAjG1M46kBiifvIhynwl'
dropbox_path = "/Prac6/aa.txt"
computer_path="aa.txt"

client = dropbox.Dropbox(dropbox_access_token)
print("[SUCCESS] dropbox account linked")

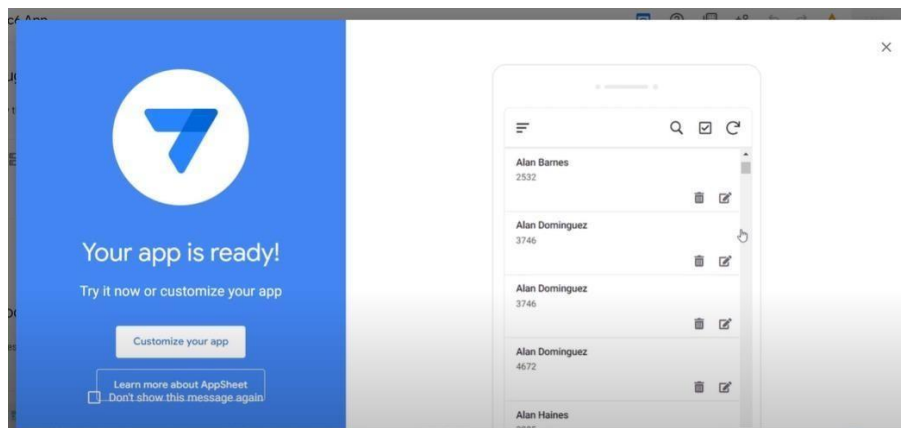
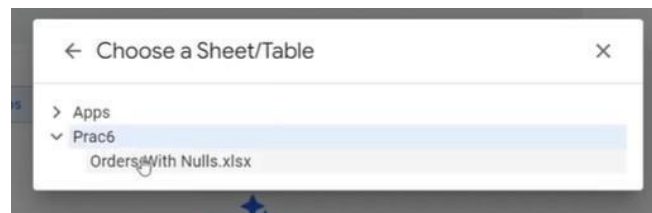
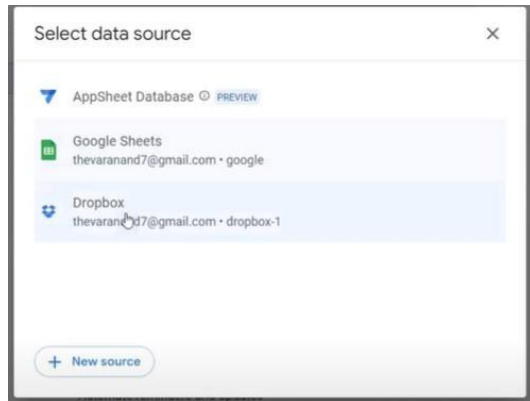
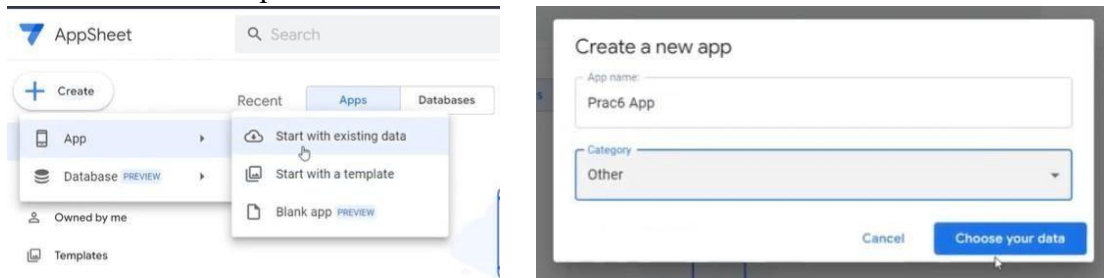
client.files_upload(open(computer_path, "rb").read(), dropbox_path)
print("[UPLOADED] {}".format(computer_path))
```

[SUCCESS] dropbox account linked
[UPLOADED] aa.txt

7. Go to <https://www.wisdomaxis.com/technology/software/data/for-reports/>, get your data sample from here and also upload the excel file in dropbox folder 'Prac6'.



8. Go to <https://about.appsheets.com/>, sign in with google. Create a new app. Choose the sample for the data.



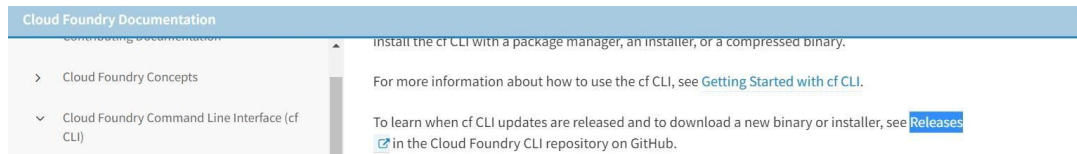
Application created.

Practical No. 7

Aim: Installing Cloud Foundry in localhost and exploring CF commands.

Implementation:

1. **Installing** Windows Powershell for CloudFoundry
<https://docs.cloudfoundry.org/cfcli/install-go-cli.html>



Download installer or binary file:

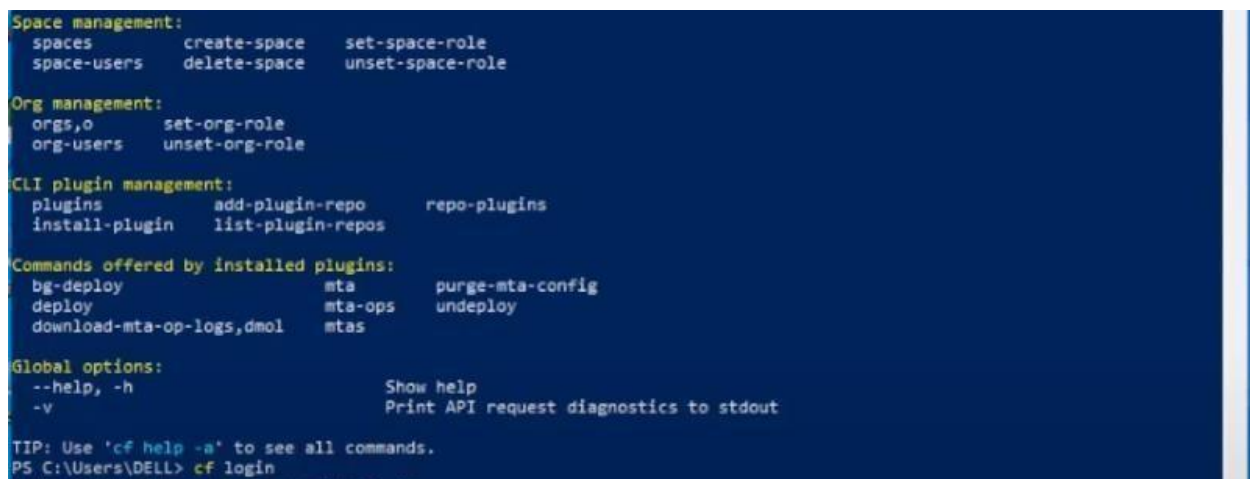
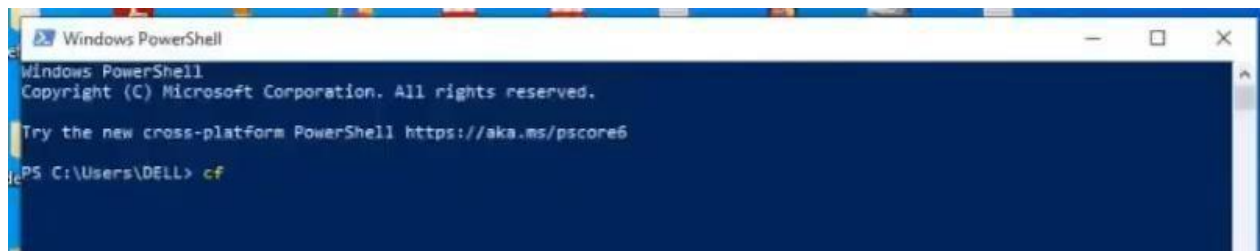
Installers

- Debian 64 bit / 32 bit (deb)
- Redhat 64 bit / 32 bit (rpm)
- macOS 64 bit / arm (pkg)
- Windows 64 bit / 32 bit (zip)

Binaries

- Linux 64 bit / 32 bit (tgz)
- macOS 64 bit / arm (tgz)
- Windows 64 bit / 32 bit (zip)

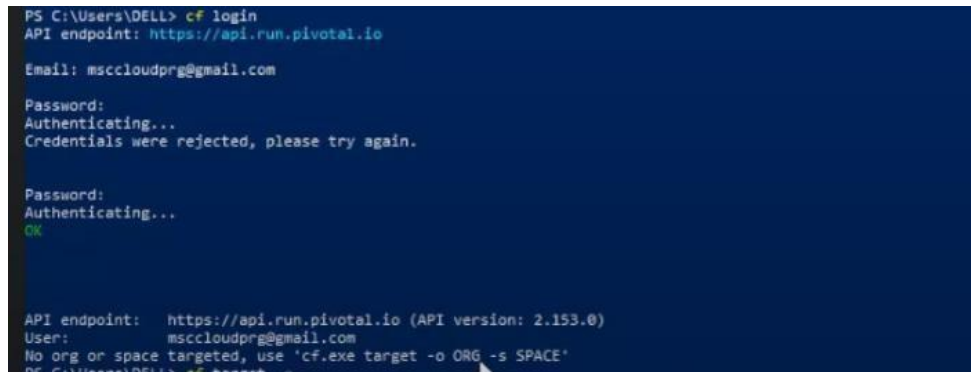
Install the file and set a system path. Then open the Windows Powershell. Type >cf to check, and if you're getting the information regarding CF, then you've successfully installed it.



Exploring Commands

2. Login with the command **>cf login**

It will have the same endpoint as IBM and we will use the same email and password as the account created in the IBM Cloud practical.



```
PS C:\Users\DELL> cf login
API endpoint: https://api.run.pivotal.io

Email: mscloudprg@gmail.com

Password:
Authenticating...
Credentials were rejected, please try again.

Password:
Authenticating...
OK

API endpoint: https://api.run.pivotal.io (API version: 2.153.0)
User: mscloudprg@gmail.com
No org or space targeted, use 'cf.exe target -o ORG -s SPACE'
PS C:\Users\DELL> cf target
```

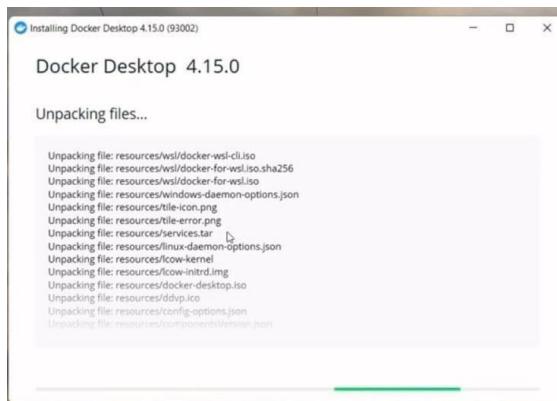
3. Generate source target using the command **>cf create-org A – Need paid version**
4. **End the practical.**

Practical No. 8

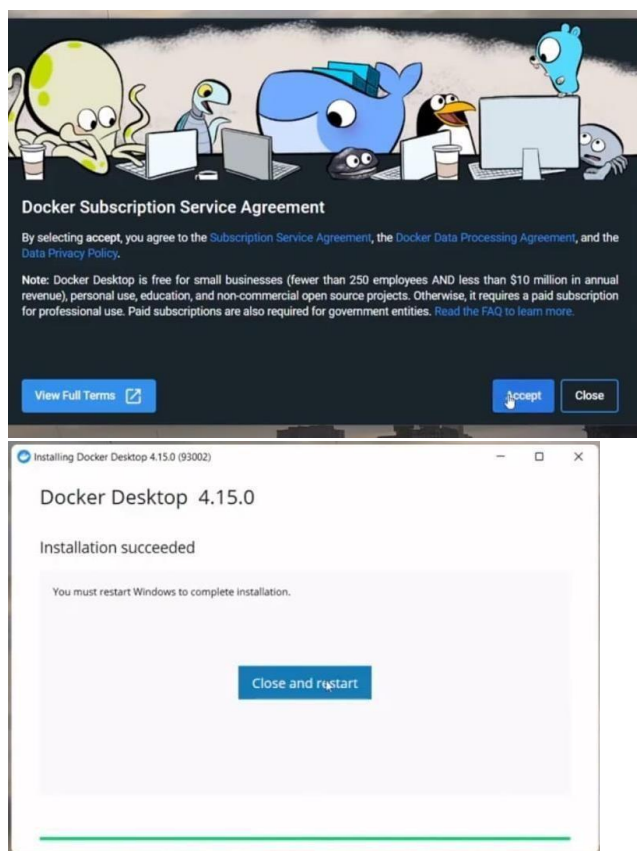
Aim: Installing and Configuring Dockers in localhost and running multiple images on a Docker Platform.

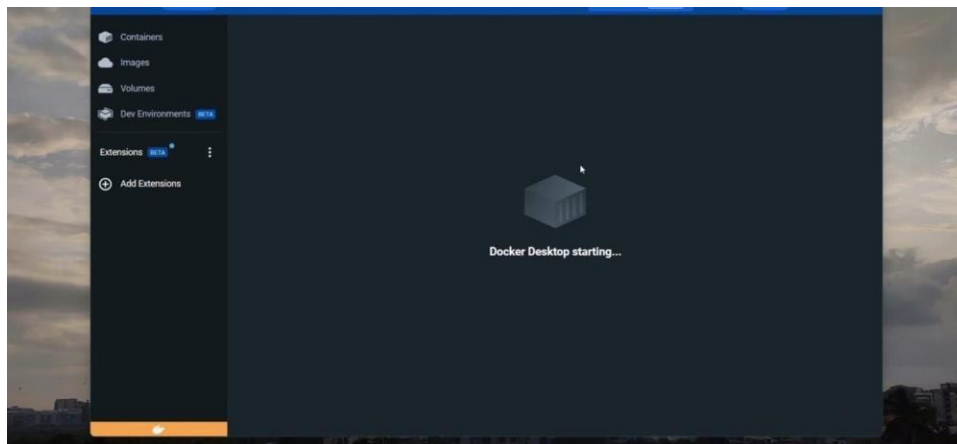
Implementation:

1. **Installing Docker Desktop** <https://www.docker.com/products/docker-desktop/> Click on Get Started and download Docker Desktop for Windows

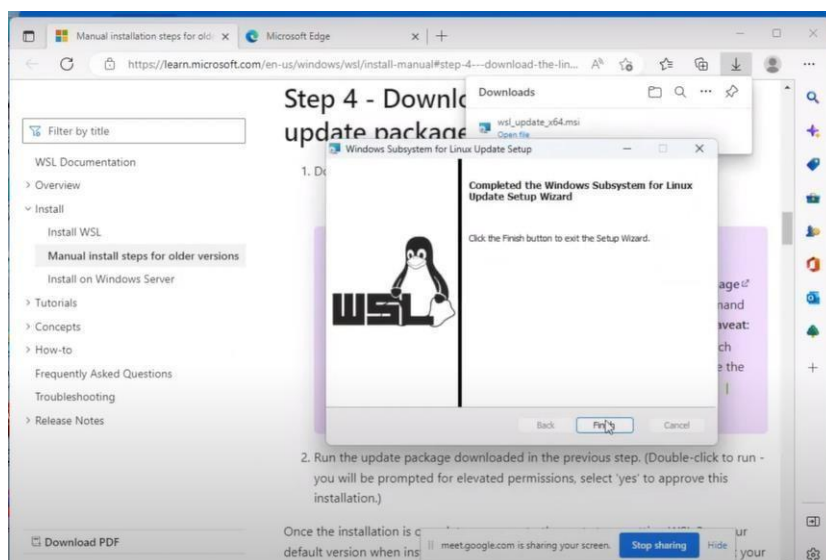
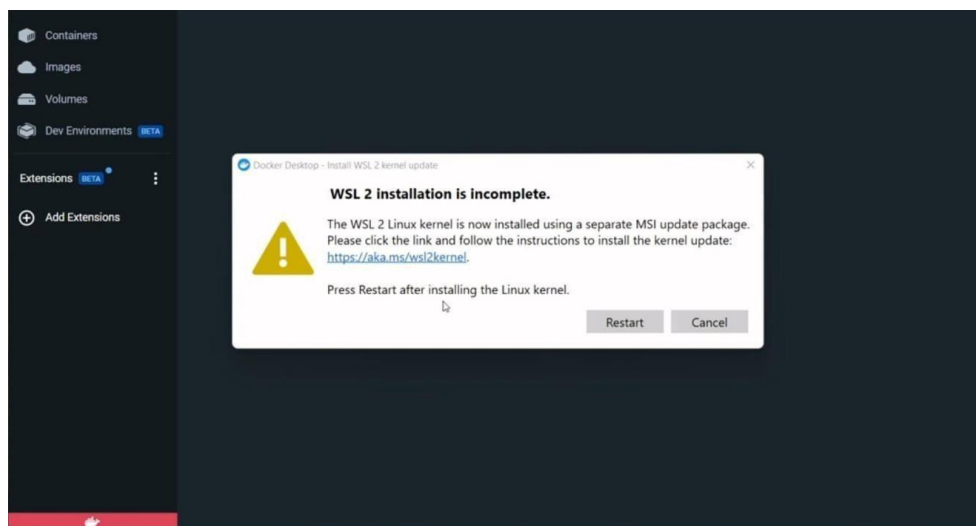


Click on Accept and complete the installation of Docker Desktop



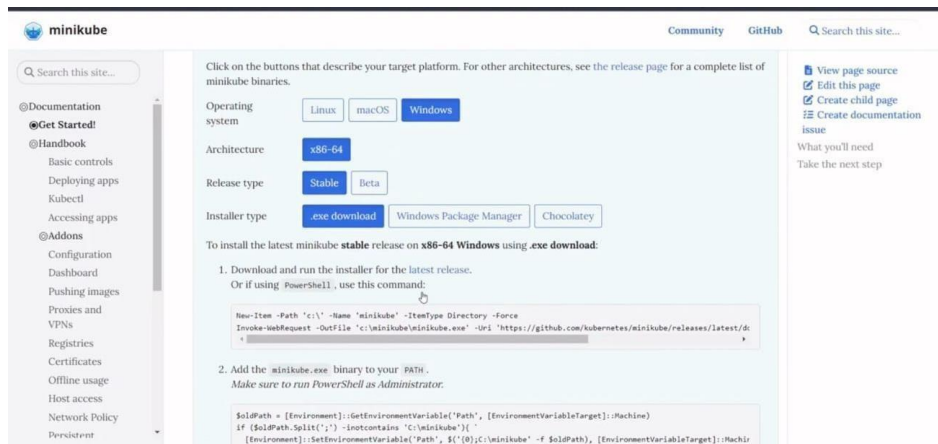


2. **Download and Install the updated version of WSL2** Click on the link and download and install the updated version



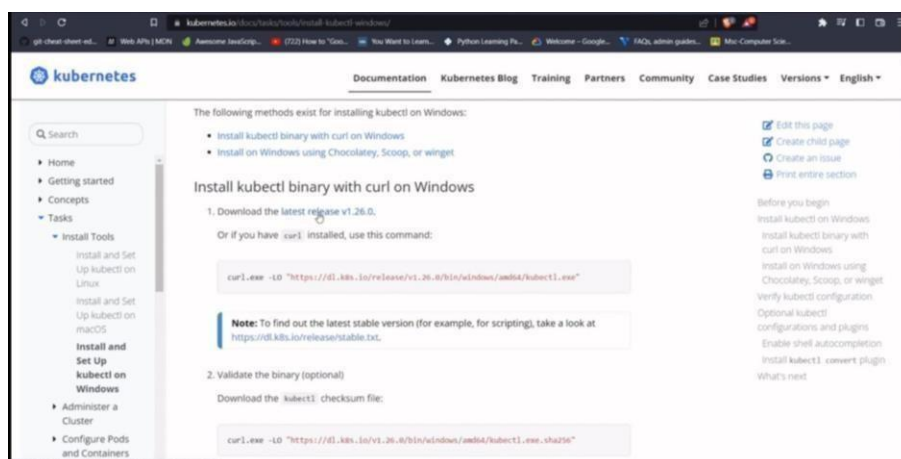
3. **Install minikube** <https://minikube.sigs.k8s.io/docs/start/>

Click on the .exe download to download minikube and Install minikube

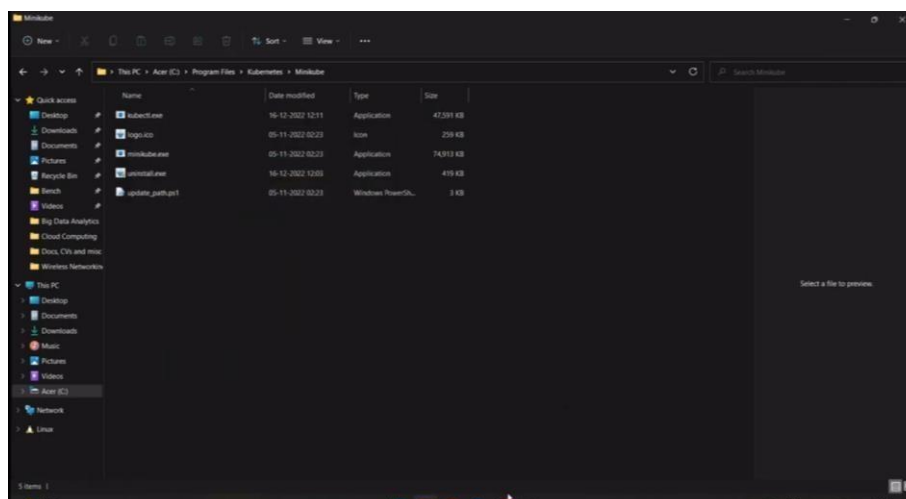


4. Install kubectl for windows <https://kubernetes.io/docs/tasks/tools/install-kubectlwindows/>

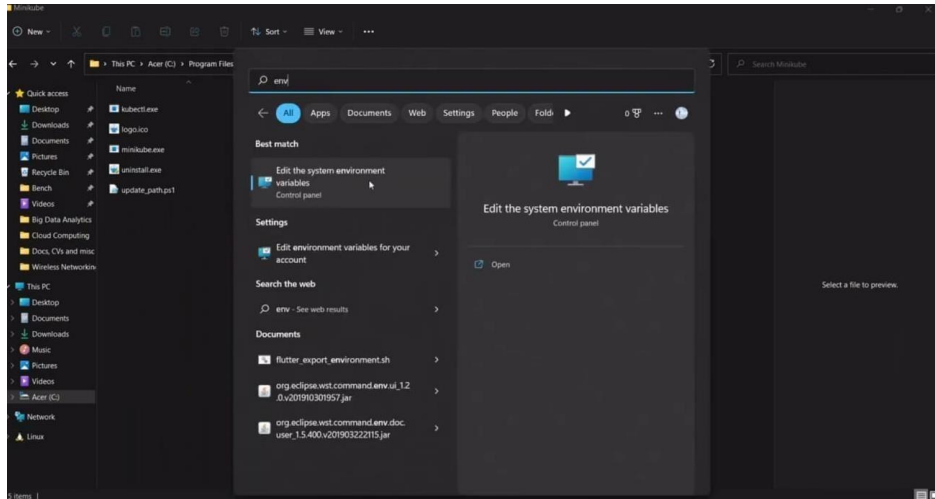
Click on the latest release and download the kubectl



Copy the kubectl where minikube is saved and then copy the path

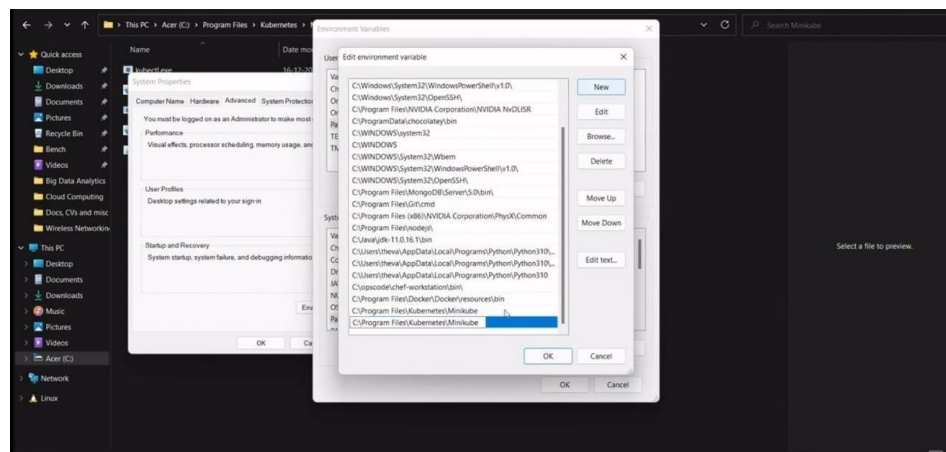
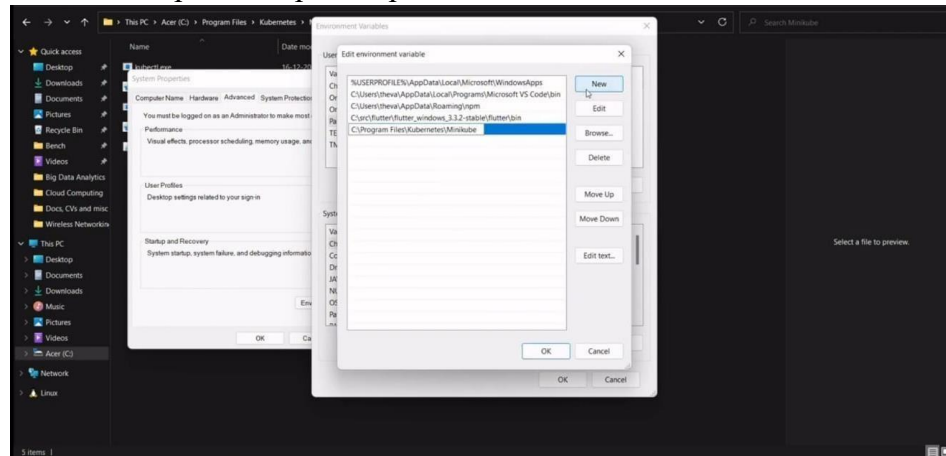


Go to environment variable:



Set the path in environment variable for the user as well as the system

Click new > paste the path copied before



Check whether Kubectl is installed

```

Windows PowerShell
auth          Inspect authorization
debug        Create debugging sessions for troubleshooting workloads and nodes

Advanced Commands:
diff          Diff the live version against a would-be applied version
apply         Apply a configuration to a resource by file name or stdin
patch         Update fields of a resource
replace       Replace a resource by file name or stdin
wait          Experimental: Wait for a specific condition on one or many resources
kustomize     Build a kustomization target from a directory or URL.

Settings Commands:
label         Update the labels on a resource
annotate      Update the annotations on a resource
completion    Output shell completion code for the specified shell (bash, zsh, fish, or powershell)

Other Commands:
alpha         Commands for features in alpha
api-resources Print the supported API resources on the server
api-versions  Print the supported API versions on the server, in the form of "group/version"
config        Modify kubeconfig files
plugin        Provides utilities for interacting with plugins
version       Print the client and server version information

Usage:

```

Open Docker and copy the line below and paste in the command prompt



```

Windows PowerShell
options      Show a list of global command-line options (applies to all commands).
license      Outputs the licenses of dependencies to a directory

Other Commands:
completion   Generate command completion for a shell

Use "minikube <command> --help" for more information about a given command.
PS C:\Users\theva> docker run -d -p 80:80 docker/getting-started
Unable to find image 'docker/getting-started:latest' locally
latest: Pulling from docker/getting-started
c158987b0551: Pull complete
1e35f6679fab: Pull complete
cb9626c74200: Pull complete
b6334b6ace34: Pull complete
f1d1c9928c82: Pull complete
9b6f639ec6ea: Pull complete
ee68d3549ec8: Pull complete
def9780e4fd4: Pull complete
277d3d14911d: Pull complete
Digest: sha256:aa5800b9692670146332b3218f73789513ddf9e4273306b491250cb86fca9499
Status: Downloaded newer image for docker/getting-started:latest
f2516de340a1a637736a114a185581eeaff40b03666a9d17ed3cde5a623e8873

```

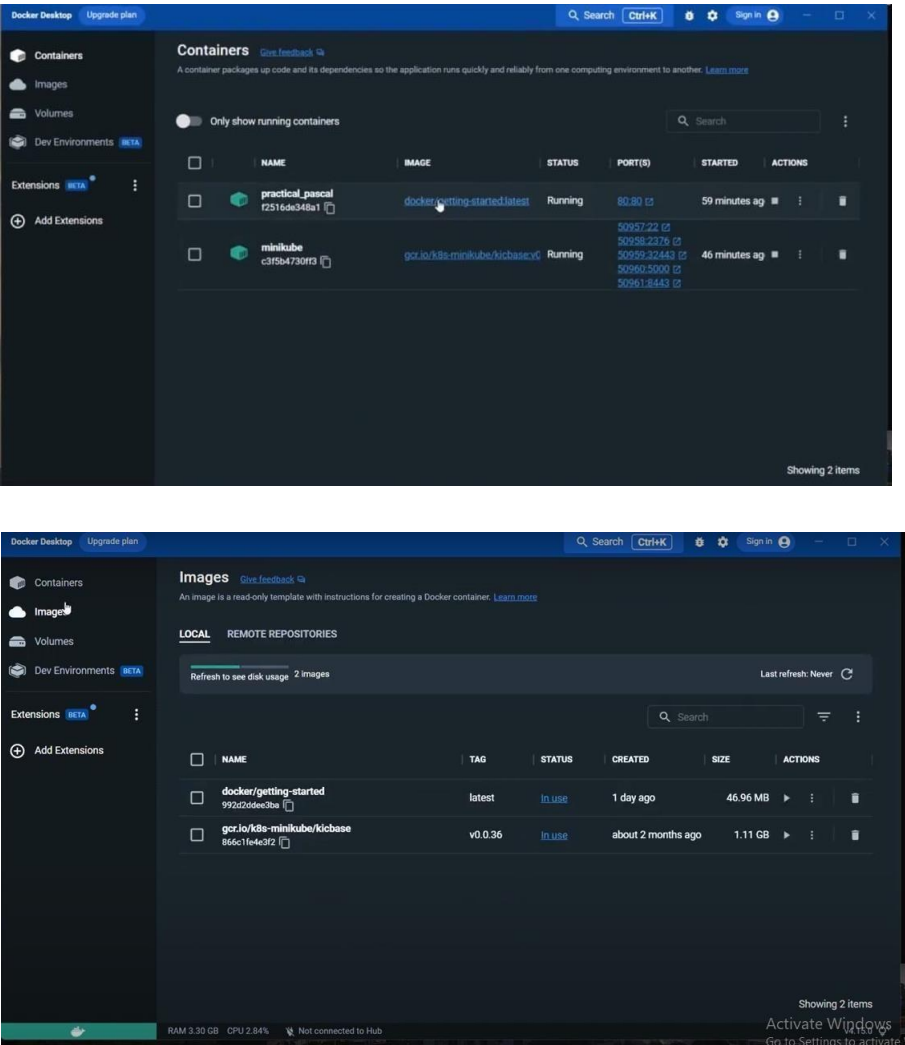
Pull the images in minikube

```

PS C:\Users\theva> minikube start --vm-driver=docker
minikube v1.28.0 on Microsoft Windows 11 Home Single Language 10.0.22000 Build 22000
Using the docker driver based on user configuration
Starting Docker Desktop driver with root privileges
Starting control plane node minikube in cluster minikube
Pulling base image ...
Downloading Kubernetes v1.25.3 preload ...
> preloaded-images-k8s-v18-v1...: 798.84 MiB / 385.44 MiB [ ] 0.20% ? p/s ?|

```

Check for the container status below



Practical No. 9

Aim: Configuring and deploying VMs/Dockers using Chef/Puppet Automation tool

Implementation:

1. To install Chef workstation, search on google for 'chef workstation windows setup' and select the first link
2. Select the windows downloads page and download the most stable version of the installer

Windows 11	22.10.1013	Architecture: x86_64 SHA256: 48049a9141ea0e0871abe279d1b2b87341232891463ee8022490a4fc15f5a1fb → Release Notes → License Information	DOWNLOAD
------------	------------	---	--------------------------

3. Run the downloaded MSI installer and once finished, open the powershell and type 'chef' on your cmd to verify

```
PS C:\Users\UDCS\Desktop> chef
The Chef command line tool for managing your infrastructure from your workstation.
Docs: https://docs.chef.io/workstation/
Patents: https://www.chef.io/patents

Usage:
  chef [command]

Available Commands:
  capture           Capture a node's state into a local chef-repo
  clean-policy-cookbooks Delete unused Policyfile cookbooks on the Chef Infra Server
  clean-policy-revisions Delete unused policy revisions on the Chef Infra Server
  completion        Generate the autocompletion script for the specified shell
  delete-policy      Delete all revisions of POLICY_NAME policy on the Chef Infra Server
  delete-policy-group Delete a policy group on Chef Infra Server
  describe-cookbook Prints cookbook checksum information for the cookbook at COOKBOOK_PATH
  diff              Generate an itemized diff of two policyfile lock documents
  env               Prints environment variables used by Chef Workstation
  exec              Runs COMMAND in the context of Chef Workstation
  export            Export a policy lock as a Chef Infra Client code repository
  gem               Runs the 'gem' command in the context of Chef Workstation's Ruby
  generate           Generate a new repository, cookbook, or other component
  help              Help about any command
  install           Install cookbooks from a policyfile and generate a locked cookbook set
  push              Push a local Policyfile lock to a policy group on the Chef Infra Server
  push-archive       Push a policy archive to a policy group on the Chef Infra Server
  report            Generate reports from a Chef Infra Server
  shell-init         Set shell context to the Chef Workstation environment
  show-policy        Show policyfile objects on the Chef Infra Server
  supermarket        chef supermarket subcommand is used to interact with cookbooks that are located in on
  c Supermarket
  undelete          Undo a delete command
  update            Updates a Policyfile.lock.json with the latest run_list and cookbooks

Flags:
  --chef-license ACCEPTANCE  Accept product license, where ACCEPTANCE is one of 'accept', 'accept-no-pers
  accept-silent'
  -c, --config CONFIG_FILE_PATH  Read configuration from CONFIG_FILE_PATH
```

4. Enter the following command
> Chef generate cookbook admin

```

PS C:\Users\UDCS\Desktop> chef generate cookbook admin
Generating cookbook admin
- Ensuring correct cookbook content
- Committing cookbook files to git

=====
Error executing action 'run' on resource 'execute[git-commit-new-files]'
=====

Vixlib::ShellOut::ShellCommandFailed
-----
Expected process to exit with [0], but received '128'
---- Begin output of git commit -m "Add generated cookbook content" ----
STDOUT:
STDERR: Author identity unknown
*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'UDCS@DESKTOP-588IHIF.(none)')
---- End output of git commit -m "Add generated cookbook content" ----
Ran git commit -m "Add generated cookbook content" returned 128

Resource Declaration:
-----
# In C:/opscode/chef-workstation/embedded/lib/ruby/gems/3.0.0/gems/chef-cli-5.6.1/lib/chef-cli

```

5. Type tree once done to see your file structure

```

PS C:\Users\UDCS\Desktop> tree
Folder PATH listing
Volume serial number is 1A80-EB15
C:.
├── .ipynb_checkpoints
├── AAKIF
├── crypt
├── demo
│   ├── .kitchen
│   │   └── logs
│   ├── compliance
│   │   ├── inputs
│   │   ├── profiles
│   │   └── waivers
│   ├── recipes
│   └── test
│       └── integration
│           └── default
├── image processing
├── IoT practical required libs
│   └── Arduino UNO Library for Proteus
├── Proteus.Pro.8.13.SP0.31525
├── saurabh
│   ├── .kitchen
│   │   └── logs
│   ├── compliance
│   │   ├── inputs
│   │   ├── profiles
│   │   └── waivers
│   ├── recipes
│   └── test
│       └── integration

```

6. Now type the following commands
 - >chef gem install kitchen-docker
7. Go to the directory of the admin cookbook and edit the kitchen.yml file as follows Driver:
 - Name:
 - docker Transport:
 - Name: docker

Platforms:

-name:

exec

Drive

r:

Name: exec

-name:exec

8. Now type the following commands

>kitchen create

>kitchen list

>kitchen converge

>kitchen list

```
PS C:\Users\UDCS\Desktop\admin> kitchen create
-----> Starting Test Kitchen (v3.3.2)
-----> Creating <default-exec>...
Finished creating <default-exec> (0m0.00s).
-----> Creating <default-exec>...
Finished creating <default-exec> (0m0.00s).
-----> Test Kitchen is finished. (0m3.23s)
PS C:\Users\UDCS\Desktop\admin> kitchen list
Instance  Driver  Provisioner  Verifier  Transport  Last Action  Last Error
default-exec Exec    ChefInfra    Inspec    Exec        Created      <None>
default-exec Exec    ChefInfra    Inspec    Exec        Created      <None>
PS C:\Users\UDCS\Desktop\admin> kitchen converge
-----> Starting Test Kitchen (v3.3.2)
-----> Converging <default-exec>...
```

```
PS C:\Users\theva\Desktop\prac10> kitchen list
C:/Users/theva/AppData/Local/chef/gem/ruby/3.0.0/gems/kitchen-docker-2.13.0/lib/docker/version
initialized constant Docker::VERSION
C:/opscode/chef-workstation/embedded/lib/ruby/gems/3.0.0/gems/docker-api-2.2.0/lib/docker/ver
s definition of VERSION was here
Instance  Driver  Provisioner  Verifier  Transport  Last Action  Last Error
default-exec Exec    ChefInfra    Inspec    Exec        Converged    <None>
default-exec Exec    ChefInfra    Inspec    Exec        Converged    <None>
```